Test-Time Compositional Generalization in Diffusion Models via Concept Discovery

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Abstract

Compositional generalization requires models to produce novel configurations from familiar parts. In diffusion models, prior compositional generation methods typically assume that the relevant concepts or conditioning signals are already available. We instead ask whether a pretrained diffusion model can discover query-specific concepts from the time-indexed scores it learns for the noisy marginals $p_t(x_t)$ and compose them at test time. Given a single out-of-distribution query, our method performs gradient ascent on $s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$ at multiple noising timesteps to recover local density modes, maps these modes into clean-space Gaussians, greedily selects relevant prototypes with a submodular likelihood objective, and combines them into a product-of-experts (PoE) teacher model with an analytic score. This teacher model can be sampled directly through classifier-free guidance or used to generate a sample pool for training a new class embedding and low-rank adapter. On held-out composition benchmarks built from ColorMNIST and CelebA, both the analytic PoE sampler and the low-rank adapted model outperform query-only and nearest trained-class baselines. These results suggest that the time-indexed score geometry of the diffusion model contains reusable density-mode concepts that support test-time compositional generation without a predefined concept library.

1 Introduction

Human intelligence is strikingly compositional. After learning familiar parts, relations, or words, people can recombine them to understand and produce novel configurations. This ability has long motivated researchers in cognitive science and linguistics. In these fields, systematic generalization is often linked to reusable primitives and operators for recombination [1, 2]. It is also a central challenge for deep learning, where neural networks may interpolate well within a training distribution but fail when familiar primitives appear in unfamiliar combinations [3, 4]. The key question is therefore whether models can not only learn useful representations, but also discover reusable units and compose them at test time. Deep learning researchers often address this problem by designing representations or architectures that support composition. For example, neural module networks assemble learned components from linguistic or program structure [5, 6]; benchmarks such as SCAN test systematic recombination [3, 4]; and visual generation methods analyze whether learned objects, attributes, or concepts can be recombined out of distribution. While these directions have made important progress, they often assume that the relevant primitives are supplied by the benchmark, labels, prompts, or a predefined concept library.

A probabilistic view offers a complementary account. Concepts can be treated as high-probability regions in an energy landscape, and composition as satisfying multiple constraints at once. This idea appears in Hopfield attractors and mixed states [7, 8], product-of-experts models (PoE) [9], and recent energy-based or diffusion composition methods [10, 11, 12]. However, these methods typically assume that the component distributions or conditioning signals are given. This leaves a gap between

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concept discovery and compositional generalization. In open-ended visual domains, the relevant primitive may depend on the query and emerge at different levels of abstraction, such as digit identity, color, texture, or facial attributes. A fuller account should therefore discover reusable concepts from the model's learned density function and compose them without fixed vocabularies or external labels.

Diffusion models provide a natural setting for this view. A pretrained diffusion model learns time-indexed scores $s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$ for the noisy marginals $p_t(x_t)$. Rather than using these scores only for reverse sampling, we repurpose them as a hierarchy of density modes: different noise levels smooth the data distribution at different scales, so local modes of p_t can act as discrete centroids at different levels of abstraction. This view is motivated by the density-mode hierarchy illustrated in Appendix A, where modes of diffusion marginals form discrete clusters that coarsen across noise levels. Starting from a single out-of-distribution query, our method maps each discovered mode into a clean-space local Gaussian, greedily selects the modes most relevant to the query with a submodular likelihood objective, and combines them into a PoE teacher model whose closed-form score can guide classifier-free diffusion sampling. We also study an optional distillation step that uses samples from the PoE to train a new class embedding and low-rank adapter, thereby absorbing the discovered composition back into the base model's score manifold. On ColorMNIST and CelebA, the discovered concept prototypes and their PoE composition outperform query-only sampling and nearest trained-class retrieval, with LoRA distillation further improving in-manifold generation.

Contributions.

- We bridge concept discovery and compositional generalization by repurposing a pretrained diffusion model as an implicit hierarchy of time-indexed density modes. Concept discovery becomes score-based mode finding over the noisy marginals $p_t(x_t)$, allowing discrete concept prototypes to be recovered at test time without a predefined primitive pool.
- We introduce a test-time composition method that greedily selects relevant discovered modes using a submodular likelihood objective and combines their clean-space local Gaussian approximations into a PoE teacher model with a closed-form score.
- On two compositional benchmarks, ColorMNIST and CelebA, we show that discovered concept prototypes and PoE composition encode richer information about unseen compositions than query-only or nearest trained-class baselines. The optional low-rank distillation step further improves in-manifold generation by learning the discovered composition back into the diffusion model.

2 Related work

Concept discovery Concept discovery aims to identify reusable abstractions between pixels and labels, such as parts, attributes, or object-level factors. Prior work has studied such concepts as interpretable directions in representation space [13, 14, 15], as discriminative prototypes [16], or as composable generative units [10, 17]. These methods reveal meaningful visual structure, but are often discriminative, rely on external supervision, or do not directly expose the hierarchy of concepts.

From a statistical view, concepts can be interpreted as local density modes, with examples assigned to their basins of attraction. This connects concept discovery to mode-seeking and hierarchical clustering methods such as mean-shift [18], density-derivative-ratio mode estimation [19], cluster trees [20], and hierarchical prototype models [21]. We instead recover concepts from the hierarchy already implicitly encoded by a pretrained diffusion model. Diffusion models admit a deep latent-variable interpretation across noise levels [22, 23, 24, 25, 26], and recent work shows that diffusion timesteps organize information from coarse semantic structure to fine detail, with phase-transition behavior at intermediate noise levels [27, 28]. This motivates our method: we discover concept prototypes by seeking modes of the smoothed marginal $p_t(x)$ using the pretrained score $\nabla_x \log p_t(x)$, requiring no additional training, external supervision, or manually specified taxonomy.

Compositional generalization Compositional generalization is the ability to recombine familiar primitives into novel configurations, a central problem in cognitive science, linguistics, and machine learning [1, 2, 3]. In AI, this ability has been studied through diagnostic benchmarks such as SCAN and gSCAN [3, 4], modular architectures that compose learned neural components [5, 6], and data-augmentation or semantic-parsing methods that encourage systematic recombination [29, 30]. These approaches differ in implementation, but typically share a common assumption: the relevant primitives are already specified by the benchmark, the language, or a predefined module set.

A complementary line of work provides a probabilistic framework for composition. Product-of-experts (PoE) models combine constraints by multiplying concept densities [9]; energy-based models use this principle to compose visual concepts [10]; and composable diffusion models implement analogous composition by adding concept-specific scores [11], with later work improving sampling through MCMC corrections [12]. Classifier-free guidance provides a related score-combination mechanism for conditional generation [31]. These methods give a mathematical account of how concepts can be composed, but still assume that the component distributions or conditioning signals are available beforehand.

Our work connects concept discovery with compositional generation, following the view that systematic generalization requires both reusable parts and an operator for recombining them [1, 2]. Closest to our setting, [17] learns a shared library of concept embeddings from unlabeled images, whereas we discover query-specific primitives at test time as modes of the unconditional diffusion density and compose their local Gaussian approximations without a predefined concept vocabulary.

Diffusion models Diffusion models learn time-indexed score fields for the noisy marginals $p_t(x)$ and are typically used as generative samplers [32, 33, 23, 34]. We repurpose this learned score geometry for concept discovery. Instead of immediately running the reverse sampler, we use $s_\theta(x,t) \approx \nabla_x \log p_t(x)$ to seek modes of intermediate marginals. This view is supported by recent analyses showing that diffusion timesteps encode structure across scales, including phase-transition behavior in which high-level class identity disappears sharply while lower-level features persist [27], suggesting that modes of p_t can serve as concept prototypes at the granularity determined by t.

3 Discovering Concepts for Product-of-Experts Composition

We study the problem of representing a query datum x_q as a composition of concept prototypes discovered from a learned base distribution. A key feature of our setting is that no predefined primitives or underlying concepts are assumed to be given. Instead, the concept prototypes are recovered from the learned distribution itself, so the method requires neither a concept library nor per-concept supervision. Once composed, the concept prototypes define an analytic product-of-experts teacher model whose score is injected into a classifier-free, compositional diffusion sampler [31, 11]. The remainder of this section develops the method in three stages. Section 3.1 introduces the notion of a concept prototype as a mode of the noisy marginals p_t, locates such modes by gradient ascent on the unconditional score, and shows that modes are clean data and can therefore be approximated by a local Gaussian in x_0-space. Building on this, Section 3.2 combines the discovered concept prototypes into a product-of-experts whose score is available in closed form. Finally, Section 3.3 plugs this score into classifier-free compositional sampling [31, 11] and describes how we distill the discovered composition into the base model with LoRA [35]. We refer readers to Figure 1 for an overview of the framework.

Notation. We work in the discrete-time diffusion setting [22, 23]. Let $x_0 \in \mathbb{R}^d$ denote a noise-free datum with distribution p_{data}, and let

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \quad t \in \{1, \dots, T\}, \tag{1}$$

be the forward-diffused state under schedule $\{\bar{\alpha}_t\}$. The noisy marginal is $p_t(x_t) = \int p_{\text{data}}(x_0)\, \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t) I)\, dx_0$, with score $\nabla_{x_t} \log p_t(x_t)$ [33, 34]. The network approximates this score through the noise-prediction head $\varepsilon_\theta(x_t, t, c)$, and the two are related by

$$s_\theta(x_t, t, c) = -\varepsilon_\theta(x_t, t, c) \big/ \sqrt{1-\bar{\alpha}_t}. \tag{2}$$

We write c for conditioning drawn from the trained set $\mathcal{C} = \{c_1, \dots, c_N\}$ and $\emptyset$ for the null token. The unseen target concept is c_*. Candidate modes are indexed by $j = 1, \dots, M$ and may come from different noise levels t_j. Each candidate has an x_t-space mode $x_t^{*,j}$, a Laplace covariance $\Sigma_j^{(t_j)}$ at that mode, and an x_0-space Gaussian expert $q_j(x_0) = \mathcal{N}(x_0; m_j, \Sigma_j)$ with $m_j = x_t^{*,j} / \sqrt{\bar{\alpha}_{t_j}}$. Greedy selection keeps K concept prototypes, indexed by a selected set $S_K \subseteq \{1, \dots, M\}$. Their weighted product-of-experts teacher is denoted q_{T}, and $c_{\text{new}} \in \mathbb{R}^{d_c}$ is the learned embedding used when this PoE teacher model is distilled into the base model using LoRA [35].

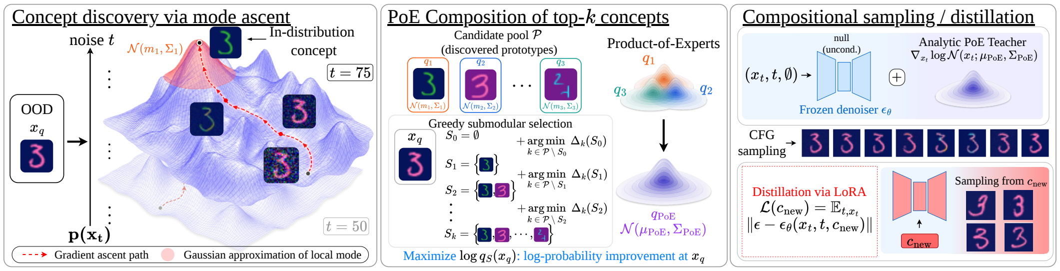

Figure 1: Overview of our DDPM-based concept discovery framework for compositional generalization, as described in Section 3.

3.1 Concept discovery via mode ascent

A diffusion model admits a deep hierarchical latent-variable interpretation in which each x_t is a latent whose prior is shaped by the learned score [22, 23, 24, 25, 26]. A line of recent work examines how this hierarchy expresses itself across the noise schedule. Sclocchi et al. [27] identify a sharp phase transition at an intermediate noise level, beyond which class identity is lost while low-level features persist, and argue that this behavior reflects a hierarchical, compositional structure in natural data. Empirically, we observe a related hierarchy in the modes of the learned noisy marginals. As the noise level increases, fine instance-level modes progressively merge into coarser concept prototypes, from which clear object-level classes emerge. See Appendix A. Taken together, these results motivate treating the modes of p_t at intermediate t as a hierarchy of concept prototypes that the pretrained score function already encodes, to be recovered from the query at test time.

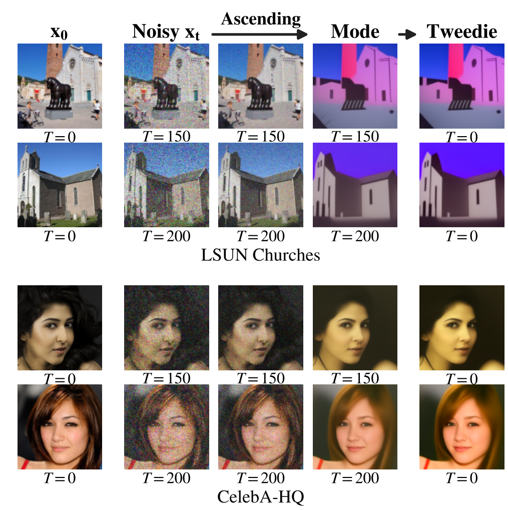

Figure 2: Found local modes in pretrained DDPMs on LSUN Churches and CelebA-HQ.

By a *mode* of p_t we mean a local maximum of the noisy marginal, i.e., a solution of

$$x_t^* \in \arg\operatorname*{local\,max}_{x_t} \; p_t(x_t). \tag{3}$$

Computing Equation (3) in closed form is intractable because p_t itself is available only implicitly through its score. We therefore locate modes by gradient ascent on the log-density,

$$x_t^{(i+1)} = x_t^{(i)} + \eta \, s_\theta\Big(x_t^{(i)}, \, t, \, \emptyset\Big), \tag{4}$$

initialized from a noised copy of the query,

$$x_t^{(0)} = \sqrt{\bar{\alpha}_t}\, x_q + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I).$$

Using the unconditional score $s_\theta(\cdot, t, \emptyset)$ ensures that ascent explores the hierarchy of the full marginal p_t rather than the conditional manifold of any trained class in $\mathcal{C}$. In practice, we replace the plain gradient step with an Adam update [36].

Modes on the rescaled data manifold. A random draw from p_t is visibly noisy by Eq. (1), yet a mode of p_t is noise-free (see Figure 2). The reason follows from Tweedie's formula [37, 38, 39, 32], which expresses the posterior mean of the clean datum as

$$\hat{x}_0(x_t) := \mathbb{E}[x_0 \mid x_t] = \frac{1}{\sqrt{\bar{\alpha}_t}} \Big(x_t + (1 - \bar{\alpha}_t)\, \nabla_{x_t} \log p_t(x_t) \Big). \tag{5}$$

At a mode, the score vanishes, so Eq. (5) reduces to $\hat{x}_0(x_t^*) = x_t^* / \sqrt{\bar{\alpha}_t}$. Conversely, by the Gaussian-convolution structure of p_t, a local mode m of p_{data} induces a nearby local mode of

p_t at approximately $\sqrt{\bar{\alpha}_t}\, m$ whenever neighboring modes are well separated relative to the kernel bandwidth $\sqrt{1-\bar{\alpha}_t}$. Combining the two gives

$$x_t^* \approx \sqrt{\bar{\alpha}_t}\, m, \qquad \hat{x}_0(x_t^*) \approx m \in \mathrm{supp}(p_{\mathrm{data}}). \tag{6}$$

Thus, unlike a random $x_t \sim p_t$, which contains a residual $\sqrt{1-\bar{\alpha}_t}\, \varepsilon$ noise component that simple rescaling cannot remove, a mode x_t^* lies at a local density peak whose noise term vanishes. We therefore treat the outputs of Eq. (4) as noise-free concept prototypes rather than noisy samples.

Local Gaussian at a mode. Having located candidate modes $\{x_t^{*,j}\}_{j=1}^M$ at possibly different noise levels $\{t_j\}_{j=1}^M$, we summarize each one as a local Gaussian expert for later composition. Because these experts are initially defined over different noisy variables x_{t_j}, we rescale them to x_0-space, obtaining experts $q_j(x_0) = \mathcal{N}(x_0; m_j, \Sigma_j)$ with $m_j = x_t^{*,j} / \sqrt{\bar{\alpha}_{t_j}}$. This common representation gives all experts the shared random variable x_0 required for product-of-experts composition, and allows each prototype to be forward-noised to any timestep needed by the sampler or LoRA distillation loss. In implementation, we use diagonal covariances for Σ_j, estimated with four Hutchinson finite-difference probes.

3.2 Product-of-experts composition

Let $\mathcal{P} = \{q_j(x_0) = \mathcal{N}(x_0; m_j, \Sigma_j)\}_{j=1}^M$ denote the candidate pool of Gaussian concept prototypes discovered by mode ascent and pulled back to x_0-space. Each candidate is now an expert over a common random variable, so the natural way to combine selected prototypes into a distribution for the out-of-distribution concept is a weighted product of experts [9]. We use per-dimension weights $w_j \in [0,1]^d$ under the diagonal Gaussian approximation, allowing different prototypes to dominate different dimension while ambiguous dimension remain softly mixed. For a selected set S_K, the product remains Gaussian, $q_{\mathrm{T}} = \mathcal{N}(\mu_{\mathrm{T}}, \Sigma_{\mathrm{T}})$, with closed-form parameters

$$q_{\mathrm{T}}(x_0) \propto \prod_{j \in S_K} q_j(x_0)^{w_j}, \quad \Sigma_{\mathrm{T}}^{-1} = \sum_{j \in S_K} \mathrm{Diag}(w_j)\, \Sigma_j^{-1}, \quad \mu_{\mathrm{T}} = \Sigma_{\mathrm{T}} \sum_{j \in S_K} \mathrm{Diag}(w_j)\, \Sigma_j^{-1} m_j. \tag{7}$$

Here exponentiation is dimension-wise, consistent with the diagonal covariance model. Combining experts in this way sharpens dimension on which selected prototypes agree while preserving uncertainty where the evidence is mixed, which is the desired behavior when a novel concept is assembled from features supported by different concept prototypes [40, 10, 11, 12].

We select the prototype set $S_K \subseteq \{1, \dots, M\}$ by greedy maximization of a per-dimension coverage objective. Write

$$\ell_{j,r}(x_q) := \log \mathcal{N}\big(x_{q,r};\, m_{j,r},\, \sigma_{j,r}^2\big)\,, \qquad r \in \{1, \dots, d\},$$

where $\Sigma_j = \mathrm{Diag}(\sigma_{j,1}^2, \dots, \sigma_{j,d}^2)$. This is the contribution of prototype j to the log-likelihood of x_q at dimension r. Under the diagonal PoE of Eq. (7), the log-likelihood decomposes per dimension. We therefore use the coverage surrogate

$$F(S) \;:=\; \sum_{r=1}^{d} \max_{j \in S} \ell_{j,r}(x_q), \tag{8}$$

which is monotone and submodular in S [41, 42]. We initialize with the best singleton, $S_1 = \{\arg\max_j F(\{j\})\}$, and then grow the set by adding the candidate with the largest marginal gain:

$$j_{i+1} \;=\; \operatorname*{arg\,max}_{j \,\in\, \{1,\dots,M\} \setminus S_i} \Delta_j(S_i), \qquad \Delta_j(S_i) \;:=\; F(S_i \cup \{j\}) - F(S_i), \qquad S_{i+1} = S_i \cup \{j_{i+1}\}, \tag{9}$$

for $i = 1, \dots, K-1$. After selection, the per-dimension composition weight of each selected prototype is the temperature-controlled softmax of its per-dimension log-likelihood over the selected set,

$$w_j(r) \;=\; \frac{\exp\big(\ell_{j,r}(x_q)/\tau\big)}{\sum_{j' \in S_K} \exp\big(\ell_{j',r}(x_q)/\tau\big)}, \qquad j \in S_K, \tag{10}$$

so that dimensions where one prototype dominates concentrate weight on that prototype, while ambiguous dimensions retain a smoother mixture.

A central consequence of the construction is that q_{T} is Gaussian in clean data space, and its score is analytic. The same is true of its forward-diffused counterpart

$$q_{\mathrm{T}}^{(t)}(x_t) = \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\mu_{\mathrm{T}},\ \bar{\alpha}_t\Sigma_{\mathrm{T}} + (1-\bar{\alpha}_t)I\big)\,,$$

whose score is

$$\nabla_{x_t} \log q_{\mathrm{T}}^{(t)}(x_t) = -\big[\bar{\alpha}_t\Sigma_{\mathrm{T}} + (1-\bar{\alpha}_t)I\big]^{-1}\big(x_t - \sqrt{\bar{\alpha}_t}\,\mu_{\mathrm{T}}\big). \tag{11}$$

3.3 Compositional sampling and distillation

Composable diffusion combines multiple concepts by adding their conditional score contributions, where each trained concept corresponds to a learned conditional score $\nabla_{x_t} \log p_t(x_t \mid c_k)$ [11]. Our discovered concept is not a trained conditioning token or class label. Instead, the selected concept prototypes define the analytic PoE teacher model q_{T}, so the score of the discovered conditional distribution is available in closed form as the score of its noisy marginal $q_{\mathrm{T}}^{(t)}$. Converting Eq. (11) to the noise-prediction parameterization via Eq. (2) gives

$$\varepsilon_{q_{\mathrm{T}}}(x_t, t) = \sqrt{1-\bar{\alpha}_t}\,\big[\bar{\alpha}_t\Sigma_{\mathrm{T}} + (1-\bar{\alpha}_t)I\big]^{-1}\big(x_t - \sqrt{\bar{\alpha}_t}\,\mu_{\mathrm{T}}\big). \tag{12}$$

We then use this analytic prediction as the conditional branch in classifier-free guidance [31]:

$$\hat{\varepsilon}(x_t, t; c_*) = \varepsilon_\theta(x_t, t, \emptyset) + w_*(t)\Big(\varepsilon_{q_{\mathrm{T}}}(x_t, t) - \varepsilon_\theta(x_t, t, \emptyset)\Big). \tag{13}$$

Thus Eq. (13) samples the discovered concept by contrasting the analytic PoE prediction against the null prediction, without requiring a trained conditional token for c_*. Equations (4)–(12) therefore form a closed framework from the query x_q to a guidance direction that the CFG/compositional sampler can consume directly.

Variance-aware guidance schedule. The residual $\varepsilon_{q_{\mathrm{T}}}(x_t, t) - \varepsilon_\theta(x_t, t, \emptyset)$ inherits the precision factor Σ_t^{-1} from Eq. (12), where $\Sigma_t := \bar{\alpha}_t\Sigma_{\mathrm{T}} + (1-\bar{\alpha}_t)I$ is the diagonal covariance of the noisy PoE marginal. A concentrated PoE distribution with small Σ_{T} can therefore produce an outsized pull at small t, while a diffuse PoE can produce a faint one. To make the guidance strength of c_* comparable across queries irrespective of prototype geometry, we set

$$w_*(t) \ = \ \min\Big(w_0 \cdot \tfrac{1}{d}\,\mathrm{tr}(\Sigma_t),\ w_{\max}\Big), \tag{14}$$

which approximately cancels the implicit Σ_t^{-1} scaling, while the cap $w_{\max}$ prevents blow-up at large t where $\Sigma_t \to I$.

Distillation with LoRA. We can further distill the PoE teacher model into the base model by jointly learning a new class embedding c_{new} and a LoRA adapter [35] attached to the frozen denoiser network. Let $\mathcal{D}_{\mathrm{T}} = \{x_0^{(i)}\}_{i=1}^{N_{\mathrm{pool}}}$ be a fixed pool obtained from the sampler of Eq. (13) with the schedule of Eq. (14), and let $\varepsilon_{\theta+\theta'}$ denote the denoiser with trainable parameters $\theta' = \{\theta_{\mathrm{LoRA}}, c_{\mathrm{new}}\}$. We optimize the standard noise-prediction loss

$$\mathcal{L}(\theta') = \mathbb{E}_{x_0\sim\mathcal{D}_{\mathrm{T}},\,t,\,\varepsilon}\left\|\varepsilon - \varepsilon_{\theta+\theta'}\big(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon,\ t,\ c_{\mathrm{new}}\big)\right\|^2, \tag{15}$$

with CFG-style dropout on c_{new} during training so that the unconditional branch remains usable at inference. Once trained, the OOD concept c_* can be generated by standard classifier-free sampling [31] conditioned on c_{new}, replacing the analytic predictor of Eq. (13). Unlike single-concept personalization formulations include textual inversion [43], DreamBooth [44], and Custom Diffusion [45], our distillation differs in that the optimization target is the analytic PoE teacher model q_{T} rather than user-provided exemplars.

Taken together, these steps close the concept discovery gap left by prior compositional-diffusion work [10, 11, 12], which assumes a given library of primitive concepts, each already represented by a trained score. Mode ascent on the pretrained unconditional score recovers concept prototypes from a single out-of-distribution query, a product-of-experts composes them into an analytic distribution q_{T} with closed-form score, and its score slots directly into the classifier-free compositional sampler. The optional LoRA distillation step then absorbs the discovered composition into the model's conditioning interface, making concept discovery and compositional generation possible from the query alone.

4 Test-Time Diffusion Compositional Generalization on OOD Query

4.1 Compositional benchmark

Test-time compositional generalization asks whether, given a single OOD query x_q depicting a held-out combination of primitive attributes, a pretrained diffusion model can sample a distribution of images of that combination, without access to any prebuilt concept library. We evaluate this on the following two compositional benchmarks:

Color-MNIST. A 32×32 RGB rendering of MNIST with three primitive factors: digit identity (10 values, 0–9), digit color (4 values from a muted high-contrast palette: yellow, green, cyan, pink), and background color (4 values: deep red, navy, dark purple, dark brown). The Cartesian product yields $10 \times 4 \times 4 = 160$ compositional slots, of which 120 are seen by the backbone during training and 40 are held out as OOD classes.

CelebA (compositional). We construct a compositional benchmark from binary CelebA [46] face attributes such as hair color, smiling, beard, and gender. Each unique realized combination of 14 such binary attributes defines a compositional class. Of the 74 realized combinations used in this experiment, 51 form the seen split for training and 23 are held out as OOD classes. Images are rendered at 128×128.

We provide additional details on both datasets in Appendix B and diffusion training in Appendix C.

4.2 Hypothesis, baselines, and metrics

Hypothesis. A single OOD query of unseen compositions x_q contains enough information for the pretrained diffusion model to expose a multi-mode neighborhood at different noising timesteps that approximates the held-out compositional class. Concretely, the discovered prototypes $\{m_j, \Sigma_j\}$ aggregated into the PoE teacher q_T of Eq. (7) should support classifier-free sampling that both (i) reconstructs the OOD concept around the query and (ii) generalizes beyond the specific query to other instances of the same OOD class. Distillation of the PoE teacher into a learned $(c_{\text{new}},\, \theta_{\text{LoRA}})$ pair (Eq. (15)) further absorbs the discovered concept onto the base model's score manifold, which we expect to yield a richer and more on-distribution conditional for sampling.

Protocol. For each OOD class, we run the full framework of Section 3 (mode ascent, PoE composition, variance-aware CFG sampling, and distillation) independently on each of the 100 queries $x_q^{(i)}$. We evaluate two configurations of our method:

- **PoE analytic CFG.** The classifier-free sampler of Eq. (13) that uses the analytic PoE score ε_{q_T} from Eq. (12) as the conditional branch, with the variance-aware schedule of Eq. (14).
- **LoRA.** We distill 256 PoE-generated samples into the base model by training a LoRA adapter on the frozen UNet together with a new class token c_{new} (Eq. (15)). At inference, we sample from the LoRA-adapted model conditioned on c_{new}.

Top-k trained classes baseline. For each query, we score every trained class $c \in \mathcal{C}$ by the average DDPM loss of x_q under that conditional,

$$\ell(c) := \mathbb{E}_{t\in\mathcal{T}_{\text{eval}},\,\varepsilon}\big\|\varepsilon - \varepsilon_\theta(\sqrt{\bar\alpha_t}x_q + \sqrt{1-\bar\alpha_t}\varepsilon,\, t,\, c)\big\|^2,$$

and pick the k smallest-loss classes $c_{(1)}, \ldots, c_{(k)}$. The top-1 baseline replaces $\varepsilon_{q_T}(x_t, t)$ in Eq. (13) with $\varepsilon_\theta(x_t, t, c_{(1)})$. The top-3 baseline composes the three smallest-loss classes via the multi-concept composition of [11] with weights $w_k = \operatorname{softmax}_k(-\ell(c_{(k)})/\tau_{\text{tk}})$. Both test whether concept discovery from the pretrained diffusion model recovers richer information about the OOD composition than retrieving or compositing the nearest seen class tokens.

Query-only (q_{x_q}) baseline. The analytic PoE teacher model is replaced with a single-query Gaussian teacher model $q_{x_q}(x_0) = \mathcal{N}(x_q, \sigma^2 I)$, where a fixed small isotropic variance is placed around the query image x_q. This teacher model is routed through the same sampler as Eq. (13). This baseline tests whether our PoE teacher model captures richer compositional structure than a distribution built only by spreading fixed variance around a single query. In contrast, the PoE teacher model combines multiple discovered concept prototypes and uses their score-implied local covariance, rather than a fixed isotropic spread around x_q.

Metrics and reference sets. For each OOD class, we evaluate generated samples using FID [47], CLIP image–image cosine similarity [48], and precision/recall in Inception-V3 feature space using the k-NN density estimator [49] with k=3. We report F1 as the harmonic mean of precision and recall. To distinguish *faithfulness to the query* from *generalization beyond the query*, we compare samples from each method (PoE, LoRA, top-1, top-3, and query-only) against two reference sets for each OOD class:

- **Faithfulness.** The 100 query images $\{x_q^{(i)}\}$ themselves. Asks whether generations concentrate near the queries that drove concept discovery.
- **Generalization.** A random set of 100 other held-out images from the same OOD class, excluding the query. This measures whether generations extend beyond the specific queries.

We refer readers to Appendix D for details on the concept discovery, sampling, and distillation configurations and hyperparameters, and to Appendix E for small-N caveats related to FID.

4.3 Results and analysis

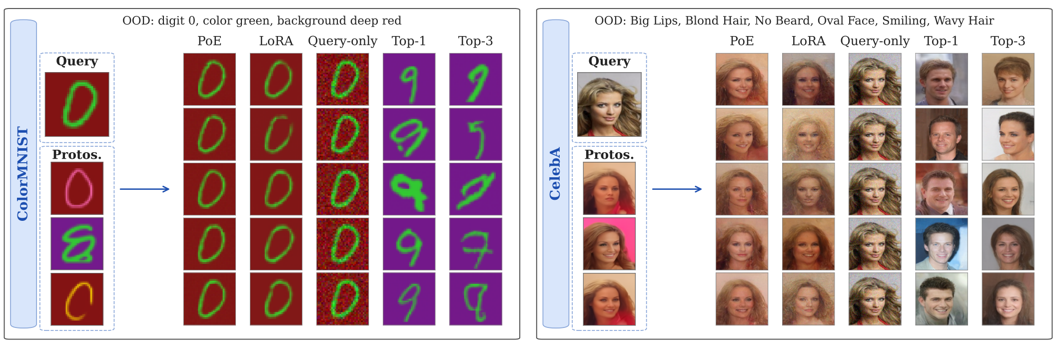

Figure 3: Examples of found prototypes, given an OOD query of unseen compositions, and generated samples from each of compared methods in ColorMNIST and CelebA dataset. We refer readers to Appendix G for additional qualitative results.

Table 1: ColorMNIST: 40 OOD compositions (mean$_{\pm \text{SE}}$). **Bold**: best; underline: 2nd.

	FID ↓		CLIP ↑ (%)		Precision ↑ (%)		Recall ↑ (%)		F1 ↑ (%)	
Method	*Faithful.*	*General.*	*Faithful.*	*General.*	*Faithful.*	*General.*	*Faithful.*	*General.*	*Faithful.*	*General.*
Top-1	$191.2_{\pm 4.8}$	$194.4_{\pm 4.6}$	$91.0_{\pm 0.2}$	$\underline{91.0}_{\pm 0.2}$	$9.5_{\pm 3.4}$	$5.3_{\pm 2.2}$	$42.2_{\pm 6.6}$	$38.8_{\pm 6.2}$	$6.2_{\pm 2.2}$	$6.5_{\pm 2.6}$
Top-3	$178.5_{\pm 5.4}$	$181.0_{\pm 5.3}$	$90.2_{\pm 0.2}$	$90.3_{\pm 0.2}$	$11.0_{\pm 3.3}$	$15.8_{\pm 3.5}$	$30.0_{\pm 5.2}$	$30.5_{\pm 5.5}$	$5.1_{\pm 1.6}$	$5.3_{\pm 1.7}$
Query-only	$247.1_{\pm 6.1}$	$254.9_{\pm 6.0}$	$82.0_{\pm 0.3}$	$81.9_{\pm 0.3}$	$0.5_{\pm 0.3}$	$0.0_{\pm 0.0}$	$2.5_{\pm 1.4}$	$1.2_{\pm 1.2}$	$0.0_{\pm 0.0}$	$0.0_{\pm 0.0}$
PoE	$\underline{81.8}_{\pm 2.6}$	$\underline{102.9}_{\pm 3.4}$	$\underline{91.2}_{\pm 0.4}$	$90.9_{\pm 0.4}$	$\underline{75.5}_{\pm 3.3}$	$\underline{67.0}_{\pm 3.4}$	$\underline{80.7}_{\pm 3.3}$	$\underline{69.8}_{\pm 3.6}$	$\underline{75.3}_{\pm 3.1}$	$\underline{63.0}_{\pm 2.7}$
+LoRA	$\mathbf{62.6}_{\pm 2.5}$	$\mathbf{84.7}_{\pm 2.8}$	$\mathbf{95.1}_{\pm 0.1}$	$\mathbf{94.8}_{\pm 0.2}$	$\mathbf{89.5}_{\pm 2.0}$	$\mathbf{82.2}_{\pm 2.7}$	$\mathbf{88.0}_{\pm 2.8}$	$\mathbf{78.8}_{\pm 3.8}$	$\mathbf{87.6}_{\pm 2.1}$	$\mathbf{76.3}_{\pm 2.6}$

Across both datasets in Tables 1 and 2, PoE and LoRA are more faithful to x_q and generalize better to other held out examples of the same OOD class. On ColorMNIST, PoE and LoRA outperform Query-only and the top-k baselines on both Faithfulness and Generalization, and LoRA further improves over PoE across the table. Its F1 remains high from Faithfulness to Generalization (87.6% and 76.3%), suggesting that distillation promotes in manifold exploration rather than memorization of the query. The ColorMNIST panel of Figure 3 explains why the trained class baselines are weak proxies for an unseen composition. The Top 1 and Top 3 columns often match surface similarity while changing the digit identity, which is consistent with their low F1 values near 5% to 7%. This supports the view that richer information about an unseen composition lies beyond nearest trained class retrieval. The Query-only column gives the complementary failure mode. Sampling around a single fixed image produces salt and pepper texture because the covariance carries little information

Table 2: CelebA: 23 OOD compositions (mean$_{\pm\text{SE}}$). **Bold**: best; underline: 2nd.

	FID ↓		CLIP ↑ (%)		Precision ↑ (%)		Recall ↑ (%)		F1 ↑ (%)	
Method	*Faithful.*	*General.*	*Faithful.*	*General.*	*Faithful.*	*General.*	*Faithful.*	*General.*	*Faithful.*	*General.*
Top-1	$193.6_{\pm2.3}$	$199.5_{\pm2.6}$	$60.4_{\pm0.4}$	$60.2_{\pm0.4}$	$39.4_{\pm4.3}$	$43.1_{\pm4.5}$	$1.4_{\pm0.6}$	$1.0_{\pm0.5}$	$1.9_{\pm0.8}$	$1.5_{\pm0.7}$
Top-3	$179.0_{\pm2.4}$	$191.9_{\pm2.7}$	$61.6_{\pm0.4}$	$61.3_{\pm0.4}$	$45.3_{\pm5.4}$	$48.9_{\pm5.2}$	$0.4_{\pm0.3}$	$0.7_{\pm0.4}$	$0.6_{\pm0.3}$	$1.0_{\pm0.6}$
Query-only	$\mathbf{107.5}_{\pm1.4}$	$\underline{147.5}_{\pm2.8}$	$66.6_{\pm0.7}$	$66.3_{\pm0.8}$	$\mathbf{80.6}_{\pm2.3}$	$48.2_{\pm3.1}$	$\mathbf{53.6}_{\pm2.9}$	$\underline{13.4}_{\pm1.3}$	$\mathbf{64.0}_{\pm2.7}$	$\underline{20.2}_{\pm1.7}$
PoE	$\underline{120.9}_{\pm3.1}$	$\mathbf{140.3}_{\pm3.9}$	$\mathbf{68.5}_{\pm0.8}$	$\mathbf{68.4}_{\pm0.9}$	$\underline{75.8}_{\pm1.4}$	$\mathbf{74.5}_{\pm1.8}$	$12.0_{\pm1.0}$	$9.0_{\pm1.0}$	$20.3_{\pm1.4}$	$15.6_{\pm1.4}$
+LoRA	$142.4_{\pm3.9}$	$160.1_{\pm4.4}$	$\underline{67.5}_{\pm0.7}$	$\underline{67.3}_{\pm0.8}$	$46.2_{\pm2.1}$	$\underline{52.7}_{\pm3.0}$	$\underline{19.9}_{\pm1.4}$	$\mathbf{16.8}_{\pm1.2}$	$\underline{26.9}_{\pm1.4}$	$\mathbf{24.7}_{\pm1.6}$

about the true class variance. By contrast, the discovered prototypes already preserve the correct digit, foreground color, and background color, and the PoE covariance uses the local score geometry around these modes to encode manifold spread.

On CelebA, the same argument becomes sharper because the images contain greater intra-class diversity and higher pixel-level information density. Query-only attains the strongest Faithfulness scores, including 64.0% F1, but its Generalization F1 drops to 20.2%. This gap indicates that q_{x_q} captures the data point more than the held-out composition. In contrast, PoE achieves the best Generalization FID, CLIP, and precision, reaching 140.3 FID, 68.4% CLIP, and 74.5% precision. LoRA further improves the balance of coverage and fidelity, reaching the highest Generalization recall and F1, with 16.8% recall and 24.7% F1 compared with Query-only at 13.4% and 20.2%. This supports the hypothesis that the PoE teacher recovers richer distributional information by composing local neighborhoods found in the score field, while LoRA distillation translates that information into a more sampleable conditional. The qualitative results in the CelebA panel of Figure 3 ground this interpretation. The PoE and LoRA columns vary identity while preserving the queried attribute combination (smiling, blond hair, wavy hair), whereas the Query-only column repeats near copies of x_q. The Top 1 and Top 3 columns drift toward different attribute combinations, showing that composition is not recovered by merging or retrieving the most relevant seen classes.

Together, the two datasets support our test-time mechanisms for using discovered modes in a frozen diffusion model, with LoRA adding a pathway for absorbing the discovered composition back into the base score manifold.

5 Conclusion, Limitation, and Future work

We presented a test-time framework that connects concept discovery and compositional generalization in pretrained diffusion models. Rather than assuming a fixed primitive library, our method uses the diffusion's learned time-indexed scores of the noisy marginals $p_t(x_t)$ to recover query-specific density modes, rescales them into clean-space local Gaussian experts, and composes them with a product-of-experts teacher model whose analytic score guides diffusion sampling. Empirically, our results support the hypothesis that unseen compositions are encoded in the local mode geometry of the learned diffusion marginals. On ColorMNIST and CelebA, PoE composition and its LoRA-distilled variant improve over query-only sampling and nearest trained-class retrieval, producing samples that better balance faithfulness to the query with generalization to other members of the same held-out composition. These findings suggest that diffusion models encode reusable concept prototypes beyond point reconstruction or class-level retrieval, providing both theoretical and empirical evidence for broader compositional applications, including language modeling [50, 51] and robotic control [52].

Limitation. A key limitation is that our protocol follows the standard compositional generalization setting: primitive factors are known, while test cases are unseen recombinations. This setting is still meaningful for large foundation models, whose pretraining may already contain a rich space of primitives, making many novel instances closer to unseen compositions than entirely unseen concepts. However, human-level intelligence also requires learning new primitives and incorporating them into future compositions [1, 2, 53]. Empirically, we observe that our framework can sometimes discover novel-looking primitives by interpolating between known primitives (see Appendix H). A full mechanism for detecting, consolidating, and reusing genuinely new primitives remains an important direction for future work.

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A Diffusion marginals as a hierarchy of modes

(a) $t = 0$ (b) $t = 50$ (c) $t = 200$ (d) $t = 300$ (e) $t = 500$

Figure 4: Modes of the noisy marginals $p_t(x_t)$ learned by a DDPM on Fashion-MNIST [54]. The $t = 0$ panel shows clean reference images, while larger t panels show prototypes recovered by mode ascent at progressively noisier marginals. As t increases, fine instance details are smoothed away and modes consolidate into coarser object-level prototypes, suggesting that diffusion marginals encode an implicit hierarchy of discrete density modes.

To motivate our use of mode finding for concept discovery, we visualize local modes of the noisy marginals $p(x_t)$ learned by a DDPM trained on Fashion-MNIST [54]. Using the mode-ascent procedure in Section 3.1, we recover concept prototypes at several noising timesteps. As t increases, the marginals become progressively smoother. Modes that initially preserve instance-level details merge into coarser object groups and eventually retain only high-level class-like structure. This supports our view that a diffusion model represents an implicit discrete hierarchy of concepts, where modes of $p(x_t)$ at different noise levels provide prototypes at different levels of abstraction.

B Additional benchmark details

Table 3: Color-MNIST primitive attributes and split.

Attribute	Cardinality
Digit identity	10
Digit color	4
Background color	4
Total compositions	160
Seen slots (training)	120
Unseen slots (OOD)	40
Per-slot images	1,000
Image format	32×32 RGB

The Color-MNIST primitive values are digit colors in {yellow, green, cyan, pink} and background colors in {deep red, navy, dark purple, dark brown}, a muted high-contrast palette chosen so that no digit color collapses against a background under low-light rendering. The 40 unseen slots are formed by holding out four (digit color, background color) pairs — (cyan, deep red), (green, navy), (pink, dark purple), (yellow, dark brown) — across all ten digits, giving $4 \times 10 = 40$ OOD slots; the remaining 120 slots train the backbone.

Table 4: CelebA compositional benchmark [46]. A compositional class is a unique combination of 14 binary CelebA attributes (Black/Blond/Brown/Gray hair, Wavy hair, No_Beard, Smiling, Big_Nose, Big_Lips, Oval_Face, Male, Sideburns, Wearing_Necklace, Bald). The 74 realized combinations used in this experiment are partitioned into a seen tier used for backbone training and a holdout tier accessible only at test time.

Attribute / split	Value
Number of binary attributes	14
Realized compositional classes	74
Seen classes (training)	51
train images	26,775
in-distribution validation images	3,825
Holdout classes (OOD)	23
images	9,914
Image format	128×128 RGB

The 23 unseen CelebA classes are chosen so that their attribute combination differs from at least one seen class on a single attribute, providing a controlled OOD setting where each held-out class is reachable from a near neighbor in the seen set. Class ids are remapped contiguously to $\{0, \ldots, 119\}$ for Color-MNIST and $\{0, \ldots, 50\}$ for CelebA before being passed to the backbone's class embedding; the backbone never sees any image from any held-out tier during training.

C Backbone architecture and pretraining

The backbone is a class-conditional `diffusers UNet2DModel` [23] trained with classifier-free guidance dropout [31] on top of two off-the-shelf UNet configurations: `google/ddpm-cifar10-32` for the 32×32 Color-MNIST backbone, and `google/ddpm-celebahq-256` (with `sample_size` overridden to 128) for the 128×128 CelebA backbone. The class-embedding table is sized to $|\mathcal{C}| + 1$ to reserve one row for the null token used during CFG dropout. Training uses the DDPM scheduler with a linear β schedule, $T = 1000$ training timesteps, ε-prediction, and an EMA on UNet weights. Color-MNIST is trained on a single GPU; CelebA is trained on 4 GPUs in bf16 mixed precision and we evaluate the EMA checkpoint at training step 500,000. Table 5 lists the per-dataset training hyperparameters; architectural fields not listed (block out-channels, layers-per-block, attention resolutions) follow the base UNet config of each dataset.

D Hyperparameters

D.1 Concept discovery hyperparameters

The per-mode covariance $\Sigma_k^{(t_k)}$ is modeled diagonal, and its diagonal is estimated from Hutchinson probes [55] of the Tweedie Jacobian. See Table 6.

D.2 Distillation hyperparameters

The trainable parameters are the LoRA factors θ_{LoRA} inserted on every linear and convolutional layer of the UNet plus the single class embedding c_{new}. The base UNet weights and the backbone's class-embedding table are frozen throughout. The training data is the fixed pool $\mathcal{D}_T = \{x_0^{(i)}\}_{i=1}^{N_{\text{pool}}}$ drawn once via the analytic compositional sampler of Eq. 13 under the variance-aware schedule of Eq. 14; each Adam step samples a mini-batch from $\mathcal{D}_T$, applies a random forward-noising at $t \sim \mathcal{U}\{1, \ldots, T\}$, and minimizes the standard ε-prediction loss of Eq. 15 jointly on $(\theta_{\text{LoRA}}, c_{\text{new}})$. CFG dropout on c_{new} at p=0.1 keeps the unconditional branch $\varepsilon_{\theta+\theta'}(x_t, t, \emptyset)$ usable at inference. Sampling along the model's score field (rather than from raw analytic Gaussian draws of q_T) keeps the distillation target on the learned image manifold. A separate $(\theta_{\text{LoRA}}, c_{\text{new}})$ pair is distilled per OOD query.

Table 5: Backbone training hyperparameters. Architectural fields not shown follow the base UNet config of each dataset (`google/ddpm-cifar10-32` or `google/ddpm-celebahq-256` with `sample_size`=128).

Parameter	Color-MNIST	CelebA
Image resolution	32	128
Channels	3	3
Base UNet config	`ddpm-cifar10-32`	`ddpm-celebahq-256` (`sample_size`=128)
Class-embedding rows (incl. null)	121	52
Optimizer	AdamW	AdamW
Learning rate	2×10^{-4}	2×10^{-4}
Weight decay	10^{-6}	10^{-6}
Gradient clip ($\|\nabla\|_2$)	1.0	1.0
Batch size	256	128 (32 per GPU $\times 4$)
Training length	100 epochs	500,000 steps
Mixed precision	bf16	bf16
EMA on UNet weights	yes	yes
Null-token dropout p	0.1	0.1
Diffusion schedule	linear β, T=1000	linear β, T=1000
Parameterization	ε-prediction	ε-prediction

Table 6: Concept discovery settings (per dataset). Both datasets use K=3 prototypes and the per-coordinate softmax weighting of Eq. (10) with temperature τ. The differences are mostly in the discovery sampling budget, which is larger on Color-MNIST because per-step ascent on a 32×32 tensor is much cheaper than on 128×128.

Hyperparameter	Color-MNIST	CelebA
Number of prototypes K	3	3
Selection objective	submodular greedy	submodular greedy
Per-coordinate softmax temperature τ	0.5	0.5
Timestep grid (`[start, end, step]`)	[50, 400, 25]	[50, 500, 50]
Ascent starts per timestep $n_{\text{per-}t}$	128	8
Ascent iterations n_{iters}	150	100
Ascent base step size	0.1	0.09
Ascent optimizer	Adam	Adam
Ascent initialization	hybrid (50% query, 50% random)	query-centered
Hutchinson probes	4	4
Hutchinson finite-difference step	5×10^{-3}	5×10^{-3}
Unconditional score source	$\emptyset$ token	$\emptyset$ token
Seed	42	42

D.3 Sampling hyperparameters

E Additional evaluation details

Small-N caveats. At N=100, FID's covariance term is poorly estimated and the metric is dominated by the squared-mean distance. The estimator is therefore biased upward but the bias is shared across methods at the same N, so within-table comparisons remain valid; absolute FID values should not be compared against published N=50,000 FIDs.

F Compute

Backbone training uses a single NVIDIA A40 GPU for ColorMNIST, completing in roughly 6 hours, and four NVIDIA RTX 6000 GPUs for CelebA, completing in roughly 50 hours. Test time concept discovery and PoE sampling both run on a single A40, and the full pipeline of section 3 (mode ascent, Hutchinson estimation of the local Hessian, submodular selection, PoE composition, and analytic CFG sampling) takes about 10-17 minutes per query.

Table 7: Distillation settings (per dataset). The trainable parameters are jointly the LoRA adapter on the frozen UNet and the new class embedding c_{new}, optimized on a fixed pool $\mathcal{D}_T$ of N_{pool} images sampled from the analytic teacher under the variance-aware schedule (Appendix D.3).

Hyperparameter	Color-MNIST	CelebA
LoRA adapter on the frozen UNet		
LoRA rank r	8	16
LoRA target modules	full UNet linear/conv	full UNet linear/conv
LoRA α	$2r$=16	$2r$=32
LoRA dropout	0.0	0.0
LoRA learning rate	1×10^{-3}	1×10^{-3}
New class embedding c_{new}		
Embedding dim d_c	matches backbone class-embed dim	matches backbone class-embed dim
Initialization	$\mathcal{N}(0,\, 0.01^2)$	$\mathcal{N}(0,\, 0.01^2)$
Learning rate	1×10^{-3}	1×10^{-3}
Optimization		
Optimizer	Adam	Adam
Mini-batch size	16	2
Epochs	10	5
CFG dropout on c_{new}	p=0.1	p=0.1
Pool $\mathcal{D}_T$ used as distillation target		
Pool size N_{pool}	256	256
Pool sampler	DDIM, η=0, 50 steps	DDIM, η=0, 50 steps
Pool guidance schedule	variance-aware (Eq. 14)	variance-aware (Eq. 14)

Table 8: Sampling settings used for both pool generation (training $\mathcal{D}_T$ for distillation, and the analytic-PoE evaluation pool) and inference from the distilled model. The variance-aware w_0, $w_{\max}$ control the strength of analytic-PoE guidance during pool generation. The LoRA sampling-w controls standard CFG strength when sampling from c_{new} after distillation; we report a sweep around the canonical value.

Hyperparameter	Color-MNIST	CelebA
DDIM scheduler	η=0, 50 steps	η=0, 50 steps
Pool size (analytic-PoE sampler)	256	256
Variance-aware base scale w_0	1.2	0.5
Variance-aware cap $w_{\max}$	2.0	2.0
Variance-mode	entropy-adaptive	entropy-adaptive
LoRA sampling-w (canonical)	0.8	1.2
LoRA sampling-w sweep	$\{0.8,\, 1.0,\, 1.2\}$	$\{0.8,\, 1.0,\, 1.2\}$
Saved gens per query per teacher	16	16
Seed (fixed across methods)	42	42

G Visualizations

We present additional visualization for both datasets in Figure 5.

H Generalization to unseen primitives.

Concept discovery operates on the score field induced by the trained DDPM, so the prototypes it can surface are confined to regions where that score field is informative. Figure 6 illustrates this boundary with two queries that share an in palette pink digit but differ in background. The first places the digit on a black background, which lies inside the convex hull of the four trained backgrounds (*deep red*, *navy*, *dark purple*, *dark brown*), all of which are dark and unsaturated. The second places the digit on a white background, which falls far outside that hull. The discovered prototypes and the PoE samples reproduce the held out black background as a coherent novel primitive, while the white query collapses onto trained dark backgrounds with the pink digit preserved but the brightness of the background lost. The discovery procedure therefore inherits the inductive biases of the backbone: it

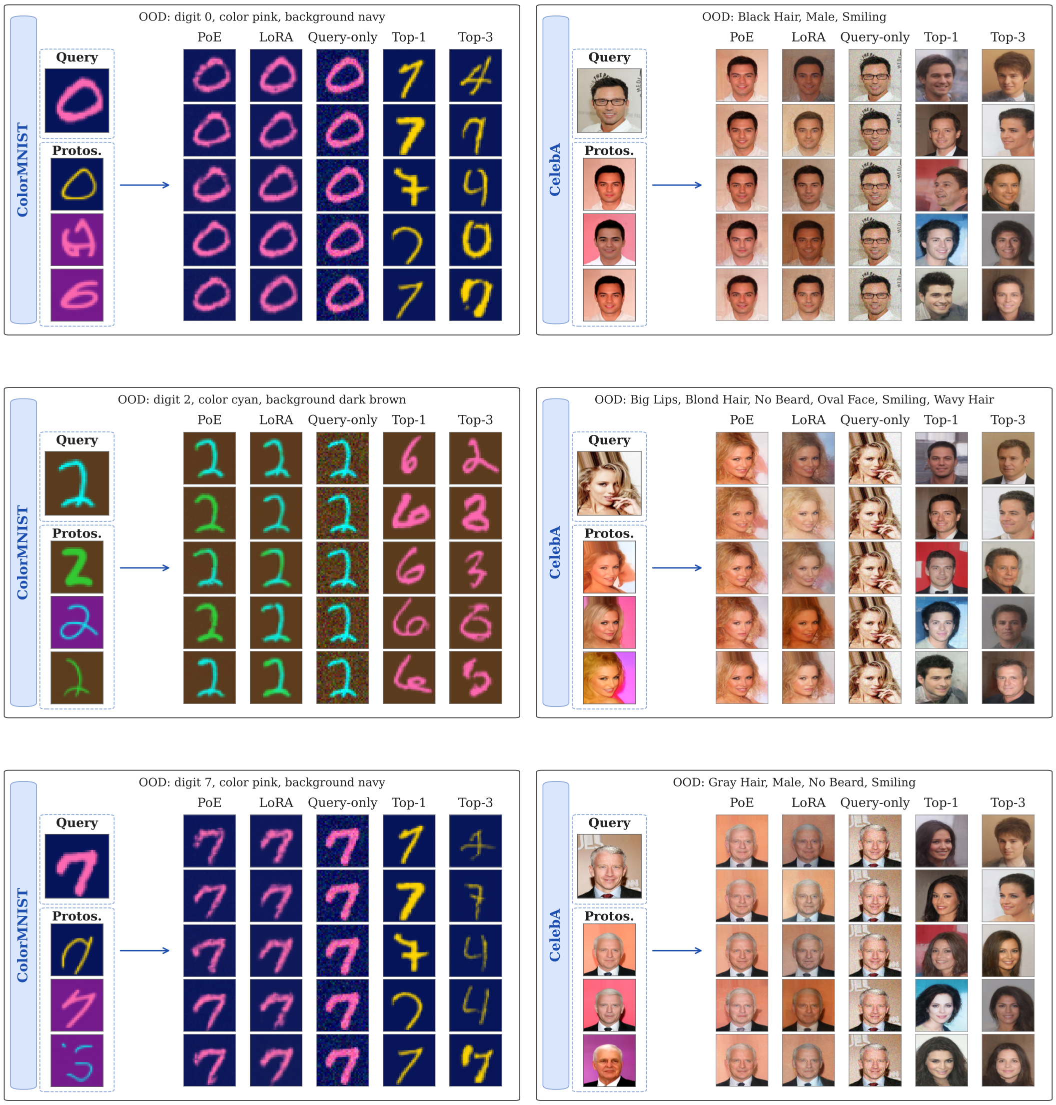

Figure 5: Additional qualitative results for the ColorMNIST and CelebA.

can compose new primitives that interpolate between seen ones, but cannot fabricate primitives that extrapolate beyond the training support.

I Broader Impacts

A potential positive impact is reducing the data and compute needed to generate novel visual concepts at test time. A potential negative impact is that the same capability could lower the barrier to generating face images with novel attribute combinations, which could be misused for misleading imagery. However, the method operates on small-scale class-conditional DDPMs and does not approach the fidelity of modern text-to-image systems.

J Licenses for Existing Assets

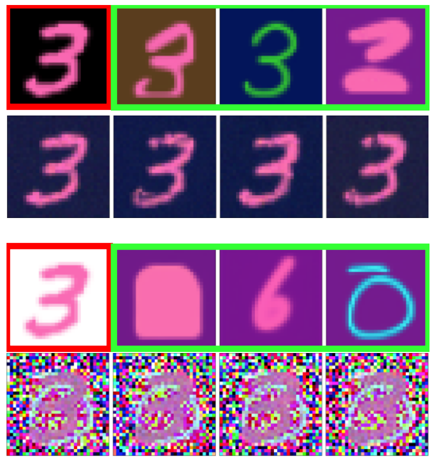

Figure 6: Concept discovery on novel background primitives, pink digit fixed from ColorMNIST. Each panel shows the query (red border), three discovered prototypes (green borders, top row), and four PoE samples (bottom row). **Top:** pink digit on a black background, an interpolation inside the convex hull of the trained backgrounds. The discovered prototypes and PoE samples reproduce the held out background as a coherent novel primitive. **Bottom:** the same digit on a white background, an extrapolation far outside the trained hull. The prototypes drift to trained dark backgrounds and the samples collapse to salt and pepper texture, evidence that the score field carries no information about the held out brightness.

Table 9: Licenses for existing assets.

Asset	License	Source
MNIST	CC BY-SA 3.0	MNIST website
CelebA	Non-commercial research	CelebA website
HF diffusers	Apache 2.0	GitHub